Computer Vision — 21CSE454T

Unit 4: Vision & Motion

Topic 2: Projection Schemes | SLO1

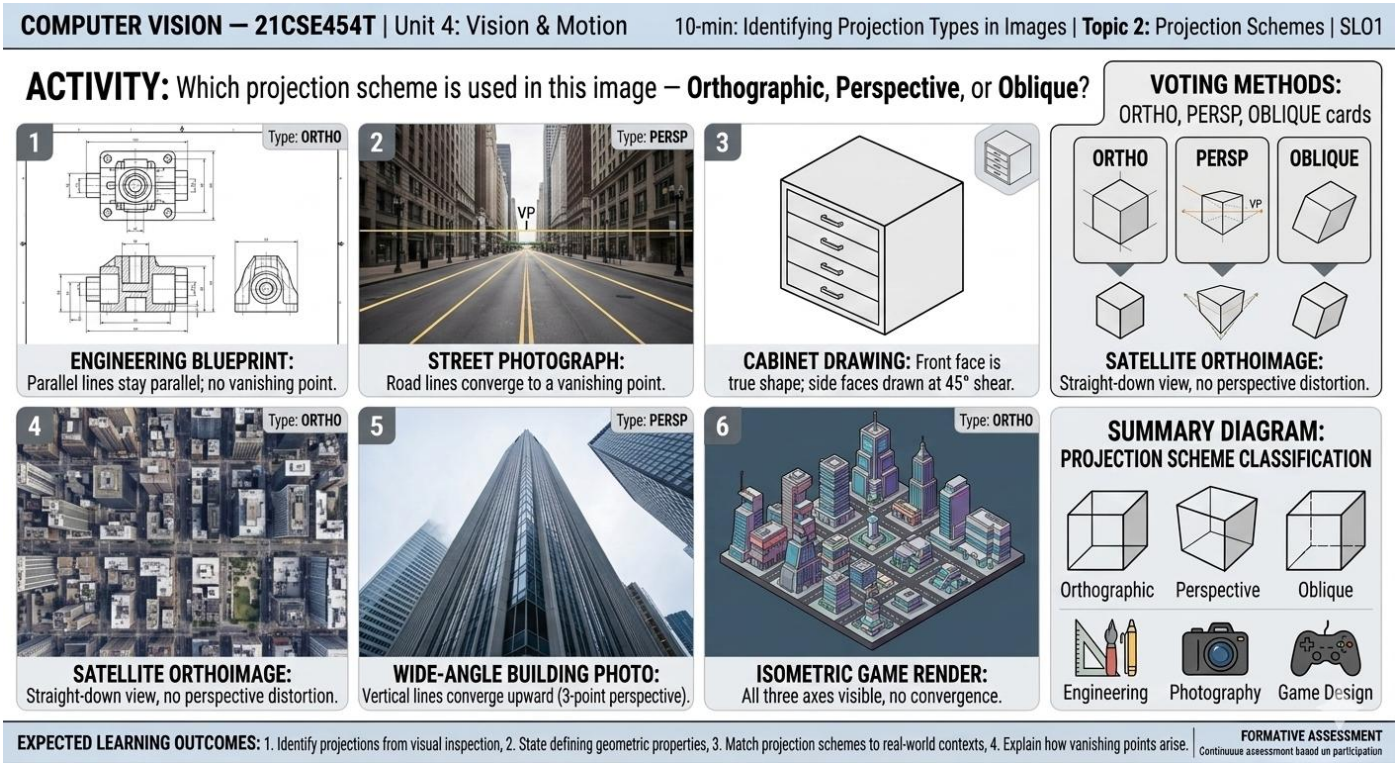
Course	Computer Vision — 21CSE454T
Unit	Unit 4: Vision & Motion
Topic	Topic 2: Projection Schemes
SLO1	Describe and differentiate between orthographic, perspective, and oblique projection schemes.
Session Title	Identifying Projection Types in Images
Duration	10 Minutes Activity
Response Method	YES / NO cards | Thumbs 👍 / 👎 | Labelled flashcards

Activity Question

"Which projection scheme is used in this image — Orthographic, Perspective, or Oblique?"

Activity Overview

Students view 6 image patches (technical drawings, photographs, architectural sketches, game renders) and identify the projection type by observing whether parallel lines converge, stay parallel, or are sheared obliquely.



Instructor Instructions

1. Display 6 images featuring different projection types — at least 2 of each: orthographic, perspective, oblique.
2. Ask students to hold up a labelled card: ORTHO / PERSP / OBLIQUE.
3. After each vote, annotate the image to show the key geometric clue (vanishing point, parallel lines, shear direction).
4. Explain the practical contexts: engineering drawings (ortho), photography (persp), cabinet drawings (oblique).
5. Conclude with a summary diagram showing all three projection types side-by-side.

Step-by-Step Activity Flow

Step	Action	Details / Expected Student Response
1	**Engineering blueprint**	Parallel lines stay parallel; no vanishing point. Expected: ORTHO. Explain preserved measurements.
2	**Street photograph**	Road lines converge to a vanishing point. Expected: PERSP. Show the horizon and VP.
3	**Cabinet drawing of a box**	Front face is true shape; side faces drawn at 45° shear. Expected: OBLIQUE.
4	**Satellite orthoimage**	Straight-down view, no perspective distortion. Expected: ORTHO.
5	**Wide-angle photo of tall building**	Vertical lines converge upward (three-point perspective). Expected: PERSP.
6	**Isometric game render**	All three axes visible, no convergence — special case of ortho. Expected: ORTHO (isometric).

Student Response Method

After each image or scenario is displayed, students respond using one of:

- Raising YES card (green) or NO card (red)
- Showing thumbs up 👍 for YES or thumbs down 👎 for NO
- Holding up labelled category cards (e.g., ORTHO / PERSP / OBLIQUE)

The instructor acknowledges the class response, reveals the correct answer, and provides a 1–2 sentence explanation before moving to the next item.

Expected Learning Outcomes

#	Learning Outcome
1	Identify orthographic, perspective, and oblique projections from visual inspection.
2	State the defining geometric property of each projection type.
3	Match projection schemes to real-world contexts (engineering, photography, illustration).
4	Explain how vanishing points arise in perspective projection.